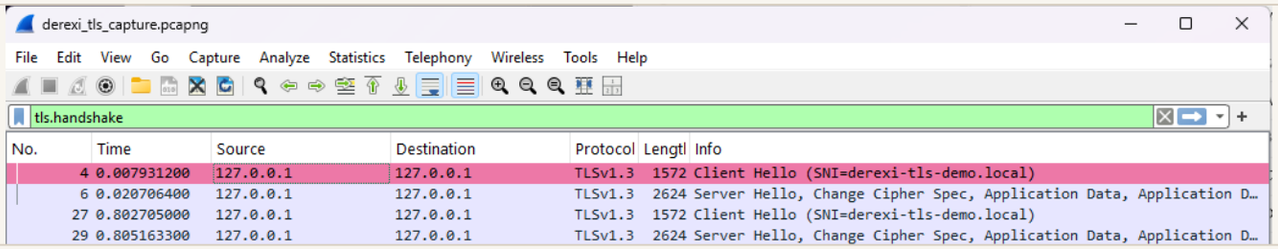
Configure cryptographic protocols

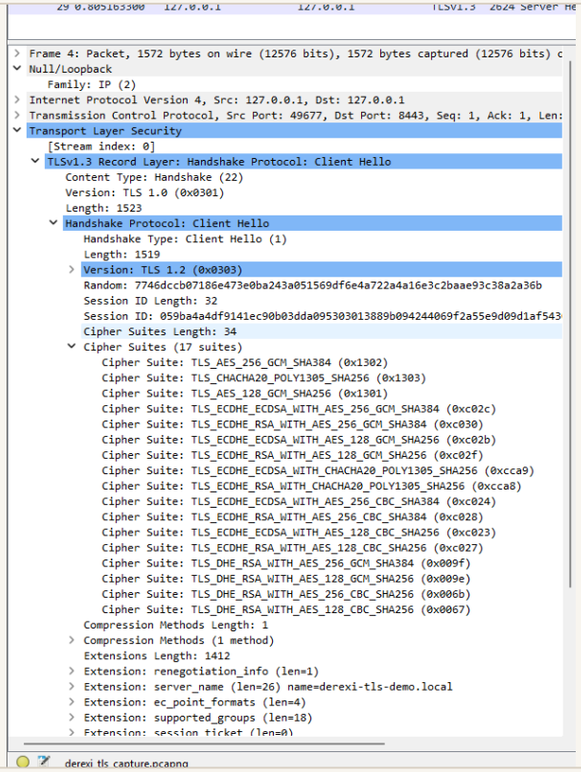
TLS 1.3 negotiation | Windows local HTTPS demonstration | 11 October 2026

OBSERVED RESULT **TLS 1.3 / TLS_AES_256_GCM_SHA384 / HTTPS 200 OK**

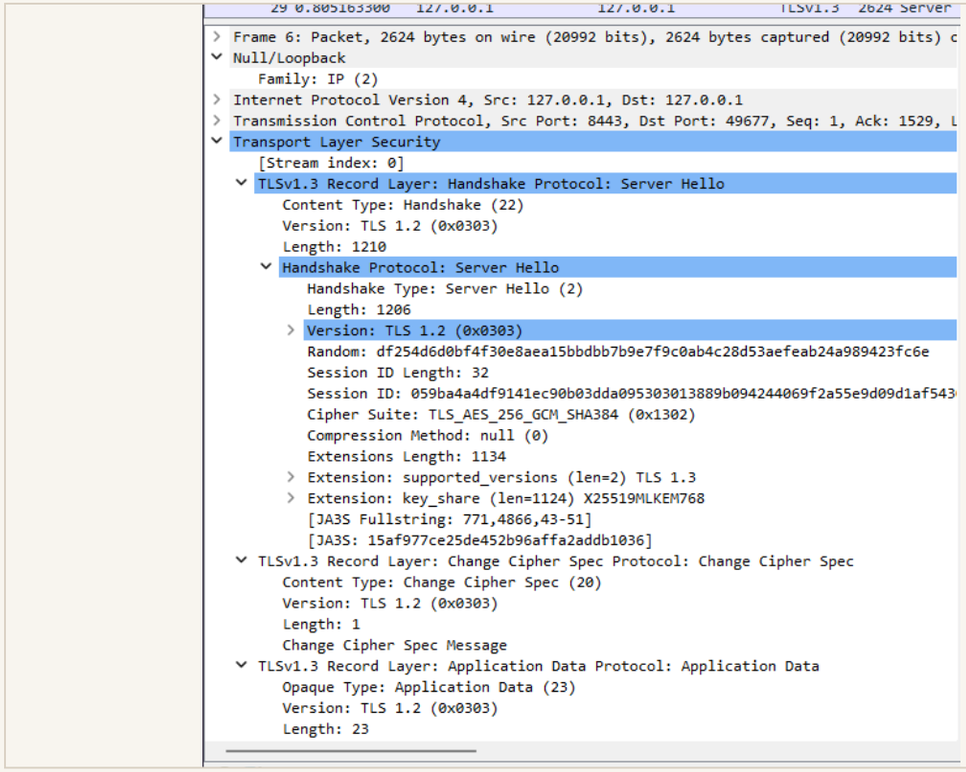
A Handshake packet overview: two Client Hello and two Server Hello packets



B Client Hello: offered cipher suites and server name



C Server Hello: negotiated suite and TLS 1.3 extension



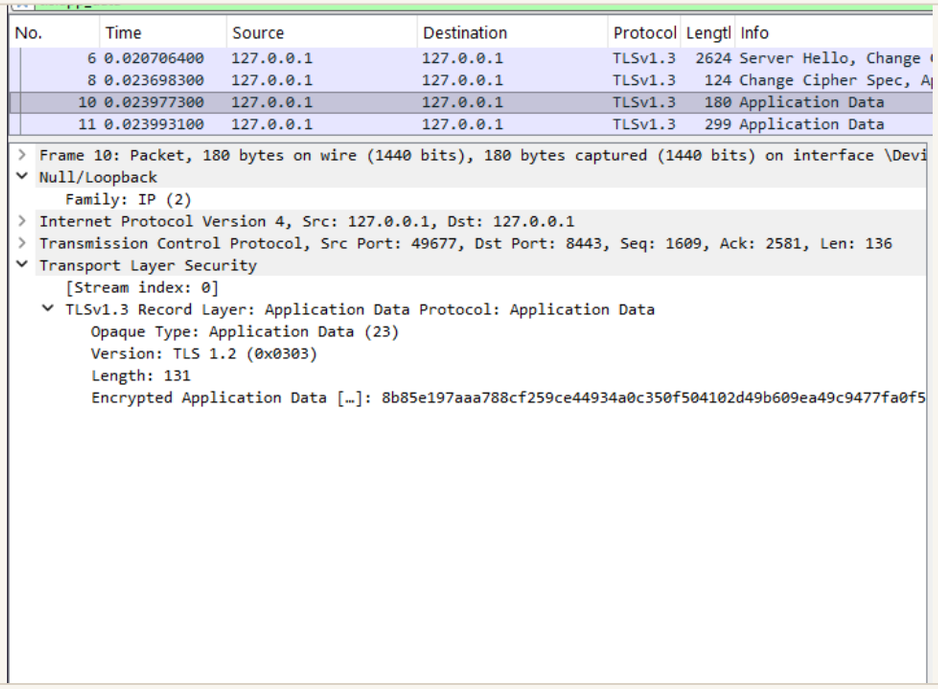
INTERPRETATION The Server Hello selected TLS_AES_256_GCM_SHA384; supported_versions identifies TLS 1.3.

The visible TLS 1.2 field is a legacy compatibility field, not the negotiated version.

Encrypted traffic & verification

Evidence-based findings | Limitations recorded transparently

D TLS Application Data: encrypted record payload observed in frame 10



TEST OUTCOME

46 packets captured

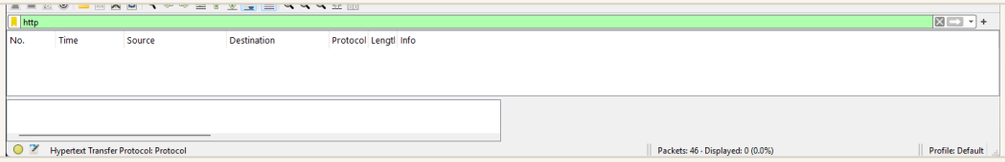
TLS version	**1.3**
Cipher suite	**AES-256-GCM**
HTTPS response	**200 OK**
Client integrity checks	**3 passed**

Captured on 127.0.0.1:8443

using a self-signed test certificate.

Local demonstration; not production TLS.

E HTTP display filter: 0 of 46 packets matched



No HTTP packets were identified by this filter; this observation alone does not prove confidentiality.

SCOPE & LIMITATIONS

TLS 1.3 encrypts certificate messages after the Server Hello; certificate subject/issuer are not visible in this capture.

SSH port 22 was closed. Keypair/fingerprint commands were simulated; no live SSH key-based login was demonstrated.

A separately verified SSH session and a secure-channel implementation may still be needed to meet the full rubric.

Evidence: five genuine Wireshark screenshots supplied by the student; summary of local tests reported by OpenCode.